James V. Raj

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Research Interests

My research is in complex systems, statistical physics, and network science. I am particularly interested in physical learning and in topological data analysis (especially persistent homology). I also study spatial networks, such as the beam networks of scale-rich metamaterials and networks that are generated from hyperuniform point patterns.

Education

University of California, Los Angeles **Sep. 2022 – June 2027**
Ph.D. in Physics

- *Advisor: Mason A. Porter*
- *Advanced to candidacy March 2025; expected completion June 2027*
- *Graduate math coursework: Optimization Theory, Calculus of Variations, Topological Data Analysis*

California Polytechnic State University, San Luis Obispo **Sep. 2017 – June 2021**
B.S. in Physics

Research Experience

Graduate Student Researcher **Oct. 2023 – present**
UCLA, Los Angeles, CA

- Advisor: Mason A. Porter, Department of Mathematics
- Showed how network architecture controls training speed across eight graph families, separating an artifact of the training budget from genuine stalls in networks with uneven degrees.
- Built a topological statistic for point clouds from persistent homology with an exact physical interpretation, and used differentiable inverse design in PyTorch to show that it detects structure that matched pair statistics miss.
- Developed a graph surrogate that predicts how stiffness varies with direction from network structure alone, replacing a finite-element solve per loading direction; on held-out networks it explains over half the variance, where an angle-only baseline explains none.
- Member of an NSF DMREF collaboration (NC State, UNC, Johns Hopkins, UCLA) on how disordered structure sets function: compared transport, curvature, and robustness in networks built from hyperuniform and random point patterns, and generated the disordered networks for 3D-printed lattices that experimental collaborators tested.

Graduate Student Researcher **June 2022 – June 2023**
UCLA, Los Angeles, CA

- Advisor: Shenshen Wang, Department of Physics and Astronomy
- Theoretical biophysics of adaptive immunity and evolving populations.

Postbaccalaureate Researcher **June 2021 – June 2022**

UC Santa Barbara, Materials Research Laboratory, Santa Barbara, CA

- Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
- Designed and ran phase-field simulations of reactive polymer blending under shear flow on UCSB's high-performance computing clusters; wrote the Python simulation-input and ParaView analysis scripts.

Summer Research Intern (NSF-funded) **June 2020 – Sep. 2020**

UC Santa Barbara, Future Leaders in Advanced Materials (FLAM) internship, Santa Barbara, CA

- Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
- Simulated polymer membrane formation on high-performance computing clusters; wrote Python tools to quantify pore structure in SEM images and compare it with simulations.

Soft Matter Research Assistant **Jan. 2019 – June 2021**

Cal Poly, San Luis Obispo, CA

- Advisors: Hilary Jacks and Nathan Keim, Physics Department
- Statistical analysis of memory formation in non-Brownian suspensions under cyclic shear, including an algorithm to read multiple stored memories from particle positions.

Nanoelectronics Research Assistant **Sep. 2018 – Aug. 2019**

Cal Poly, San Luis Obispo, CA

- Advisor: Colleen Marlow, Physics Department; includes a 2019 summer research internship
- Simulations of carbon nanotube network field-effect transistors: network morphology, percolation, and sensitivity to electrostatic gating.

Publications

Peer-reviewed journal articles

1. R. Sengupta, M. D. Tikekar, **J. V. Raj**, K. T. Delaney, M. C. Villet, and G. H. Fredrickson, "Phase-field simulations of morphology development in reactive polymer blending," *J. Rheol.* **67**, 1 (2023). doi:10.1122/8.0000523

Preprints and papers under review

2. **J. V. Raj**, X. Sun, C. E. Maher, K. A. Newhall, and M. A. Porter, "Local geometric and transport properties of networks that are generated from hyperuniform point patterns," arXiv:2511.21082 (2025); in revision at *Phys. Rev. E*. Code: github.com/DMREF-networks/HuPPI-Network-Analysis.

Manuscripts in preparation

3. **J. V. Raj**, L. Gyllingberg, P. B. Pietrzkowski, M. Stern, and M. A. Porter, "Network structure controls how quickly local learning opens spectral gaps in spring networks" (2026).
4. **J. V. Raj**, C. E. Maher, K. A. Newhall, and M. A. Porter, "Size-resolved H_0 persistence as an exact coalescence observable" (2026).
5. **J. V. Raj**, C. E. Maher, K. A. Newhall, and M. A. Porter, "A directional transport surrogate for stiffness in disordered beam networks" (2026).

Presentations

1. **[Contributed Talk] J. V. Raj**, L. Gyllingberg, P. B. Pietrzkowski, M. Stern, and M. A. Porter, "Network structure shapes accessible solutions in physical learning," Conference on Complex Systems (CCS 2026), Binghamton, NY, Oct. 2026.
2. **[Contributed Talk]** K. E. Daniels, S. Gat, S. Kondaveeti, C. E. Maher, M. A. Porter, **J. V. Raj**, S. Ramirez, and C. Rock, "Characterizing 3D-printed disordered metamaterials under static and oscillatory driving," APS Global Physics Summit, Denver, CO, Mar. 2026. (Presented by K. E. Daniels.)
3. **[Contributed Talk] J. V. Raj** and M. A. Porter, "Topological and transport properties of spatially embedded networks generated by hyperuniform point patterns," APS March Meeting, Anaheim, CA, Mar. 2025.
4. **[Contributed Talk] J. V. Raj**, J. Garcia, and G. H. Fredrickson, "Understanding how the glass transition concentration affects polymer membrane microstructure," Future Leaders in Advanced Materials Summer Symposium, UC Santa Barbara, Santa Barbara, CA, Aug. 2020.
5. **[Poster] J. V. Raj**, R. Valenzuela, and C. Marlow, "Electrical current characteristics of simulated carbon nanotube network field-effect transistors," APS Far West Section Meeting, Stanford University, Stanford, CA, Nov. 2019.

Teaching

UCLA (Teaching Assistant)

- Physics 1B: Oscillations, Waves, Electric and Magnetic Fields (Fall 2024, Spring 2025)
- Physics 1A: Mechanics (Winter 2025)
- Astronomy 115: Introduction to Galactic Astronomy (Winter 2023, Winter 2024)
- Physics 32: Mathematical Methods (Fall 2023)
- Physics 112: Thermal Physics (Spring 2023)

Mentoring and Outreach

- Research mentor to Pola B. Pietrzkowski (UCLA undergraduate) on how network topology affects physical learning (2026–present)
- Research co-mentor, with Mason A. Porter, to Emma Vega (UCLA undergraduate) on percolation in Erdős–Rényi, Barabási–Albert, and configuration-model networks (2025–present)
- Research co-mentor, with Mason A. Porter, to Xiaohan Sun (UCLA undergraduate) on curvature and effective resistance in networks generated from point patterns; coauthor of arXiv:2511.21082 (2024–2025)
- Mentor to eight undergraduate students, Critical MASS mentorship program, UCLA Marginalized Identities in Physics & Astronomy (MiPA) (2023–2026)
- Mentor to two incoming Ph.D. students, UCLA Physics Ph.D. Mentorship Program (2023–2026)

Honors and Awards

- Outstanding Teaching Award (twice), UCLA Physics and Astronomy (2022–2023 and 2023–2024)
- Future Leaders in Advanced Materials Internship Award (2020)
- William and Linda Frost Undergraduate Research Award, Cal Poly (2019)

Service

- Assisted with a referee report for *Interfaces and Free Boundaries* (2025)
- Administrator of the Porter research group's shared Linux compute server (2025–present)

Memberships

- American Physical Society (Member, 2018–present)
- Sigma Pi Sigma, physics honor society, Cal Poly (Vice President, 2020–2021)